



**BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
INSTRUCTION DIVISION**

FIRST SEMESTER 2026-2027

Course Handout (Part - II)

Date: 1st August, 2026

In addition to part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : PHA F312
Course Title : Medicinal Chemistry-I
Instructor-in-Charge : R. Mahesh
Instructors : for laboratory component - as in timetable

1. Scope and Objective of the Course:

This course deals with the study of important classes of drugs that are listed below. Various aspects like structure, properties, therapeutic and pharmaceutical importance and the uses of drug molecules of both natural and synthetic origin will be covered. Study of physiochemical properties, mechanism of action, S.A.R. and metabolism of drugs dealt hereunder will also be emphasized. Special emphasis will be laid on drug discovery, drug design and related topics.

2. Text Book :

1. William O Foye- "Principles of Medicinal Chemistry", Sixth Edition, Lippincott Williams and Wilkinson, Philadelphia, 2008

3. Reference Books :

1. Robert F Doerge-" Wilson and Gisvold's Textbook of Organic Medicinal and Pharmaceutical Chemistry", Ninth Edition, J.B. Lippincott Company, Phil.,1991.
2. Remington's Pharmaceutical Sciences, 18th ed.,1990.
3. Introduction to Medicinal Chemistry by Graham L Patrick. Oxford University press, 5th Ed., 2013
4. Vogel's text book of Practical Organic Chemistry, Longman, 5th Ed., 1996

Note: It is strongly recommended that the students follow the latest edition of the textbook and not too old, earlier versions of the textbook as much information has been added/deleted based on latest developments and understanding of drugs, drug-receptor interactions as well their pharmacology.



4. Course Plan:

Lec. No.	Topic to be Covered	Learning Objectives	Ref.
1-6	Medicinal Chemistry –outline of concepts in drug discovery, drug action	Physiochemical properties of drug molecules, structural features and pharmacological activity, Drug discovery design and development, drug receptor interactions, etc.	TB-1-5, RB, other sources
7-10	Cholinergic drugs	Design and chemistry of cholinergics, anti-cholinergics, and anti-cholinesterases	TBI-12, RB.4-22 TB2-23
11-13	Adrenergic drugs	Biosynthesis, metabolism, SAR, and synthesis of related drugs .	TBI-13, RB.4-23 TB2-22
14-15	Local anesthetics and general anesthetics	Mode of Action, Syntheses, SAR	TBI-18, TB2-6
16-18	Sedatives and hypnotics, Anti-anxiety agents	Mode of Action, Syntheses, SAR	TBI-19,21-22 TB2-7
19- 21	Anti-epileptics	Mode of Action, Syntheses, SAR	TBI-20,TB2-8
22-24	Anti-psychotics	Mode of Action, Syntheses, SAR	TBI-21,22,TB2-9
25-28	Anti histaminic drugs (anti-allergic, anti-ulcer)	Mode of Action, Syntheses, SAR of anti-allergic and antiulcer drugs	TBI-37,TB2-14
29-30	Thyroid and Antithyroid drugs	Mode of Action, Syntheses, SAR	TBI-34
31-32	Drugs for erectile dysfunction	Structure, synthesis, etc.	TBI-33
32-34	Opioid analgesics	Mode of Action, Syntheses, SAR	TBI-24,TB2-11
35-39	NSAIDs, related drugs	Mode of Action, Syntheses, SAR	TBI-36,TB2-12

5. Practicum experience:

Name of Experiment	No of practical hours	References/ chapters
Synthesis of various medicinal compounds	As in timetable	Different chapters in text book/ practical book(s)

Note: Practicals / Practicum experience involves training a student in the synthesis of few drug molecules including learning of various processes involved such as purification, crystallization, adapting protocols to better yield of compounds and processes involved therein. Hence it is important that students pay strong attention to details including safety instructions and judiciously perform experiments as per instructions provided by lab. in-charges. Being an exposure by learning of principles taught in theory by practice, this practicum experience is very important and failure to participate and perform as required, may be considered as incompleteness of course work requirements towards obtaining a valid course grade, as all aspects of practicum experiences are evaluated.



6. Evaluation:

	Evaluative Component	Weightage (in %)	Duration	Date and Time	Comments
1	Mid. Sem. Test \$\$\$	25	90 mins	As announced by AUGSD/AGSRD	Closed book
2	Continuous Assessment of classroom participation and discussions and resulting surprise quiz(zes)#	10	continuous	As announced in class by I/C	Open Book & or Closed Book
2	Continuous Assessment (lab. component) , Lab. comprehensive examination components (Practicum experience)**	30	continuous	As announced in class by I/C	Open Book & or Closed Book
3	Comprehensive Examination	35	180 mins.	As announced by AUGSD/AGSRD	Open Book & or Closed Book

\$\$\$ Mid Sem. test may be replaced by quizzes and or class test(s).

* Continuous assessment may include quiz(zes)#, Practicum experience** Laboratory - Day to Day work and quiz(zes), Viva-Voce, Home assignment, and Lab. component including Lab. Compre., etc.

** Practicum experience, Assignment(s) may be practical / theory oriented, for which a proper documentation and submission of records and reports in a standard format should be submitted as per deadline(s) that would be announced. It may also include a viva and or seminar presentation(s). No make-ups will be allowed for evaluation components items * and ** above including surprise quiz(es) and class assignments. Anyone not present in class during such evaluatives will be considered as ABSENT and no marks will be given.

It is strongly advised that all students prepare their own class notes and relevant information from text, reference material, as in handouts would only be allowed for consultation during assessments of open book/notes. Photocopies of any material, written or printed will not be permitted. Stapled sheets , loose sheets of information written or printed, photocopies of slides used for discussion in class will not be allowed for evaluatives unless otherwise specified. No other text/reference material (including edition mentioned in handout) will be permitted for reference during open book assessments.

Students are also requested to refresh their knowledge in basic organic reactions and physiology.



It is strongly advised that all students **prepare their own class notes** and relevant information from text, reference material, as in handouts would only be allowed for consultation during assessments of open book/notes. **Photocopies of any material, written or printed will not be permitted.** **Stapled sheets, loose sheets of information written or printed, photocopies of slides used for discussion in class, will not be allowed for evaluatives, unless otherwise specified. Someone else's notes will also not be permitted. No other text/reference material (including edition mentioned in handout) will be permitted for reference during open book assessments.**

* Slides used during class hours provide key information for which additional supportive information is expected to be collected from sources aforementioned. These slides will **not be shared through mail** and hence students are requested/advised to make their own notes during class hours. Recent developments in the area/topic will be discussed in class based on their significance to healthcare delivery and hence some information on therapeutic benefits and toxicity effects, besides others, **may differ (not present) from the information in text, reference material** and hence students are expected to take note of such key discussions during contact hours. **Such discussions held in class will be considered as primary source of information in assessments.** Also note that all classes will be held (online/in person) as announced by AUGRD/AUGSD, from time to time and students are requested to note the same.

Mid-Sem. Grading would be announced, once at least 30-40 % evaluation components are completed.

For all evaluation components, information given in class lectures, tutorials, text books and reference books, in the same order will be considered, as correct. All evaluation components are equally important, irrespective of weightage. Hence, students failing to attend or absenting themselves in one or many of the evaluation components, including participation in classroom discussion, active participation in laboratory sessions may become ineligible for obtaining a valid grade, at the end of the semester. Attendance in lectures, tutorials, other evaluatives like quiz(zes) and practicum experience, are all equally important as they are all integral components of learning, **irrespective of weightage** and may be taken into consideration, during grading.

Students are strongly advised to keep away from absenting themselves from all aforementioned contact sessions. Clearing the course would require adequate performance in written quizzes, tests**, examinations**, and in practicum experience components, separately (i.e. procuring low marks in evaluation components, aforementioned**, other than practicum experience, would not suffice, to clear the course). All announcements regarding the course will be mostly discussed / shared during class/lab. contact hours and no mail may be sent to individual student. It is the responsibility of the student to be aware of all discussions that happen during lecture/tutorial/lab. contact sessions.**

Any other adaptive changes in the handout, will be announced in class, if any.

Reading Assignments: Students are advised to read, collect additional information on the above mentioned topics as per given schedule.

7. Chamber consultation hours: (3170-S); Time to be announced in class, based on mutual free time.

8. Notices: Notices concerning the course will be displayed on the Pharmacy Department Notice Board only.



4. Make-Ups: Make-Ups are not given as a routine. It is solely dependent upon the GENUINENESS OF THE CIRCUMSTANCES under which a student fails to appear in a scheduled evaluation component. In such circumstances, prior permission should be obtained from the Instructor-in- Charge. In no case the make-up letter be slipped inside the chamber of the Instructor-In-Charge or intimation sent after the evaluative (applicable only for mid-sem.test and comprehensive examination). The decision of the Instructor- in-Charge in these aspects will be final. No messaging or mails will be entertained as students being residential on campus, can met the I/C of the course, in terms of any difficulty.

Instructor - in -Charge
PHA F312

Table of contents in Foye's principles of medicinal chemistry (Text Material)

Historical Perspective of Medicinal Chemistry

Part I: Principles of Drug Discovery

- | Chapter 1. Drug Discovery from Natural Products
- | Chapter 2. DrugDesign and Relationship of Functional Groups to Pharmacologic Activity
- | Chapter 3. Molecular Modeling and in Silico
- | Chapter 4. Receptors and Drug Action
- | Chapter 5. Drug Discovery Through Enzyme Inhibition
- | Chapter 6. Pharmaceutical Biotechnology-From Nucleic Acids to Personalized Medicine
- | Chapter 7. Peptide and Protein Drugs
- | Chapter 8. Antisense Therapeutic Agents
- | Chapter 9. Physicochemical and Biopharmaceutical Properties of Drug Substances and Pharmacokinetics
- | Chapter 10. Drug Metabolism
- | Chapter 11. Drug Regulation: An Overview

Part II: Drug Receptors Affecting Neurotransmission and Enzymes as Catalytic Receptors

- | Overview of Drug Receptors: APerspective
- | Chapter 12. Drugs Affecting CholinergicNeurotransmission
- | Chapter 13. Adrenergic Receptors and Drugs Affecting Adrenergic Neurotransmission
- | Chapter 14. Serotonin Receptors and Drugs Affecting Serotonergic Neurotransmission
- | Chapter 15. Amino Acid Neurotransmitters in the Central Nervous System
- | Chapter 16. Inhibitors of Nerve Conduction: Local Anesthetics
- | Chapter 17. Phosphodiesterase Inhibitors

Part III: Pharmacodynamic Agents

Section 1: Drugs Affecting Central Nervous System



- | Chapter 18. General Anesthetics
- | Chapter 19. Sedative-Hypnotics
- | Chapter 20. Antiseizure Agents

- | Chapter 21. Antidepressants
- | Chapter 22. Psychotherapeutic Drugs: Antipsychotic and Anxiolytic Agents
- | Chapter 23. Hallucinogens, Stimulants and Related Drugs of Abuse
- | Chapter 24. Opioid Analgesics
- | Chapter 25. Drugs Used to Treat Neuromuscular Disorders: Antiparkinsonian and Spasmolytic Agents

Section 2: Drugs Affecting the Cardiovascular System

- | Chapter 26. Cardiac Agents: Cardiac Glycosides, Antianginal, and Antiarrhythmic Drugs
- | Chapter 27. Diuretics
- | Chapter 28. Angiotensin Converting Enzyme Inhibitors, Antagonists and Calcium Blockers
- | Chapter 29. Central and Peripheral Sympatholytics and Vasodilators
- | Chapter 30. Antihyperlipoproteinemics and Inhibitors of Cholesterol Biosynthesis
- | Chapter 31. Antithrombotics, Thrombolytics, Coagulants, and Plasma Extenders

Section 3: Drugs Affecting the Hormonal Systems

- | Chapter 32. Insulin and Drugs Used for the Treatment of Diabetes
- | Chapter 33. Adrenocorticoids
- | Chapter 34. Thyroid Function and Thyroid Drugs
- | Chapter 35. Calcium Homeostasis

Section 4: Drugs Affecting the Immune Systems

- | Chapter 36. Nonsteroidal Anti-Inflammatory Drugs
- | Chapter 37. Antihistamines and Related Antiallergic and Antiulcer Agents

Section 5: Chemotherapeutic Agents

- | Chapter 38. Antibiotics and Antimicrobial Agents
- | Chapter 39. Antiparasitic Agents
- | Chapter 40. Antifungal Agents
- | Chapter 41. Antimycobacterial Agents
- | Chapter 42. Cancer and Chemotherapy
- | Chapter 43. Antiviral Agents and Protease Inhibitors

Part IV: Disease State Management

- | Chapter 44. Asthma and Chronic Obstructive Pulmonary Disease
- | Chapter 45. Men's Health
- | Chapter 46. Women's Health

Appendix A: pKa Values for Some Drugs and Miscellaneous Organic Acids and Bases

Appendix B: pH Values for Tissue Fluids